

1.2 HW. Write-up.

15.

P	Q	R	$(P \wedge Q)$	$(P \wedge Q) \vee R$
T	T	T	T	T
T	T	F	T	T
T	F	T	F	T
T	F	F	F	F
F	T	T	F	T
F	T	F	F	F
F	F	T	F	T
F	F	F	F	F

$(P \wedge Q) \vee R$ is a contingency
since it is true for at least
one assignment of values for the
variables and false for others.

36.

P	Q	$P \rightarrow Q$	$(\neg P) \rightarrow (\neg Q)$
T	T	T	T
T	F	F	T
F	T	T	F
F	F	T	T

$P \rightarrow Q \not\equiv (\neg P) \rightarrow (\neg Q)$ since
they have different truth columns

37.

P	Q	$P \rightarrow Q$	$Q \rightarrow P$
T	T	T	T
T	F	F	T
F	T	T	F
F	F	T	T

$P \rightarrow Q \not\equiv Q \rightarrow P$ since they
have different truth columns

62. $f_{\leftrightarrow}(f_{\neg}(T), f_{\wedge}(T, T)) = f_{\leftrightarrow}(F, T) = F$